

Turn this assignment in on gradescope. All work must be neat and legible. Turn in a "final draft". There should be nothing scratched out and your work should flow generally from left to right and top to bottom.

(Yes - there are only two problems!)

1. Find two non-trivial solutions to $x^2 - 60y^2 = 1$. (Use the Chakravala algorithm; you don't need to show the arithmetic, but you do need to show the steps in the algorithm. That is, the (x, y, k) in each step).

Since at each step of the algorithm, we want to minimize $|m^2 - N|$, we start with $m = 8$ since $8^2 = 64$ is closest to 60, and get the first triple $(8, 1, 4)$. For our first improvement, we need to find the next m such that $\frac{x+ym}{k}$ is an integer, while still minimizing $|m^2 - 60|$. We see that $m \equiv 0 \pmod{4}$, and again, we have $m = 8$ is the closest to $\sqrt{60}$. Our next x is

$$x_{\text{new}} = \frac{xm + 60y}{|k|} = \frac{8 * 8 + 60 * 1}{4} = 31.$$

Our next y is

$$y_{\text{new}} = \frac{x + ym}{|k|} = \frac{8 + 1 * 8}{4} = 4.$$

Our next k is

$$k_{\text{new}} = \frac{m^2 - 60}{k} = \frac{64 - 60}{4} = 1.$$

Thus, we have found one solution, namely $(31, 4, 1)$. Now, we can pick a different starting m for our first step. We can pick $m = 7$ since $7^2 = 49$ is the second closest to 60, and get the first triple $(7, 1, -12)$. I will omit the arithmetic and

2. Using the algorithm with starting values $(10, 1, 8)$ find a solution to $x^2 - 92y^2 = 1$. Once you get to the point where $k = \pm 4$, use some reduction to get to a solution (note: I showed how to do this when $k = \pm 2$; do something similar here). Show your work as I mentioned above.